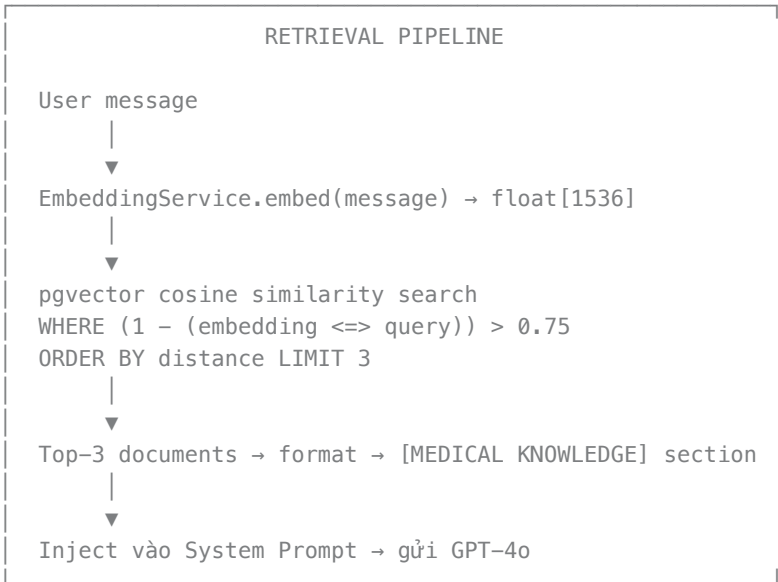
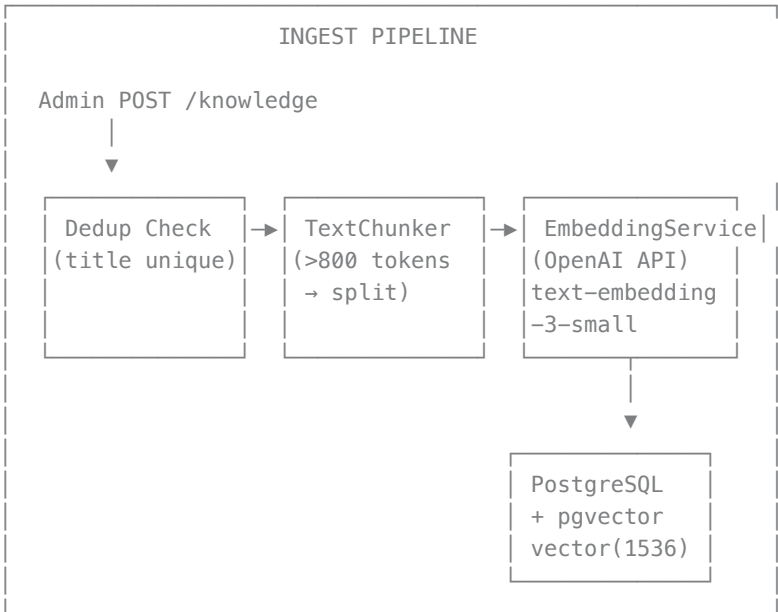


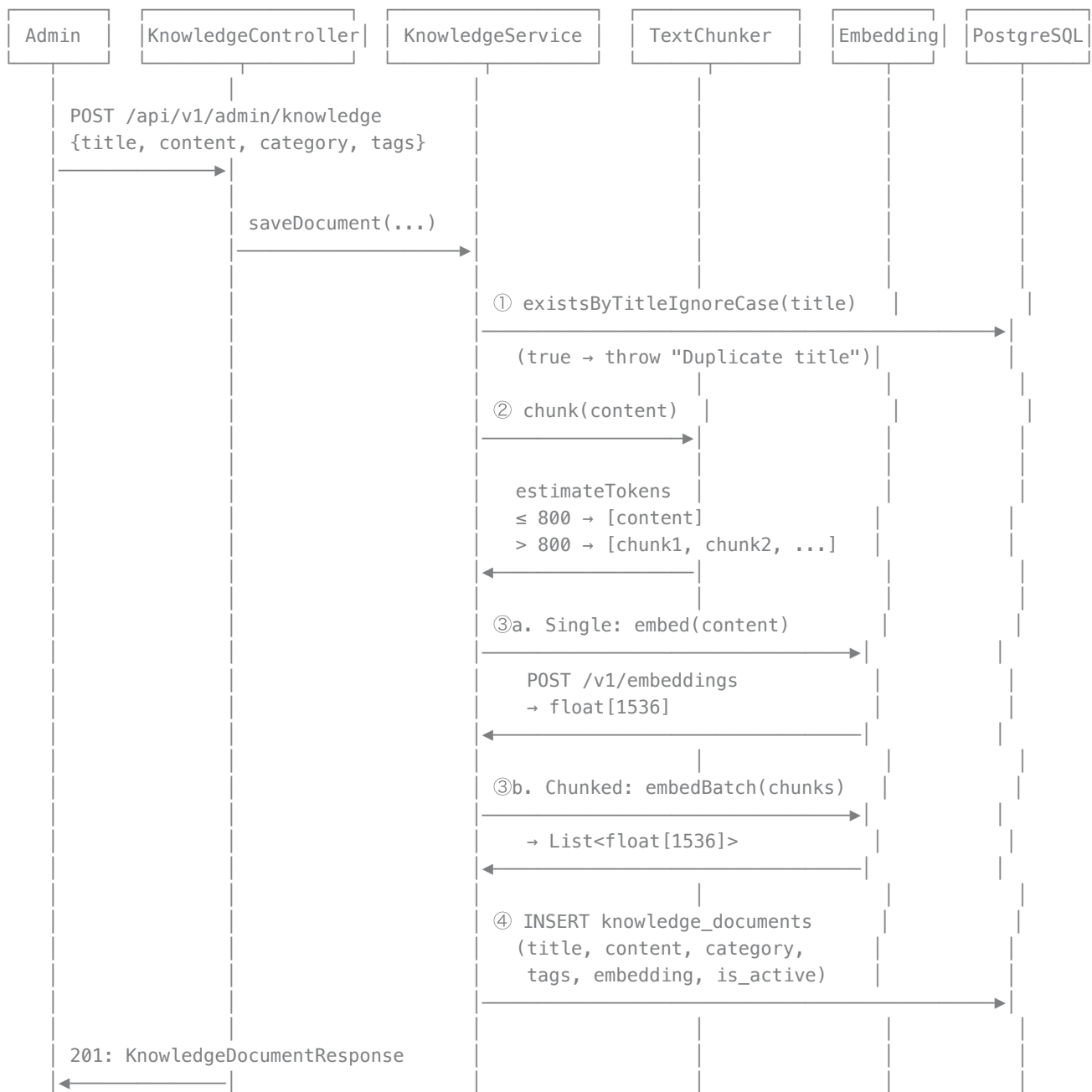
Luồng 4: Build RAG — Knowledge Base Management

Mô tả chi tiết luồng import, embedding, chunking, search và quản lý knowledge documents cho hệ thống RAG.

4.1 Tổng quan kiến trúc RAG



4.2 Sequence Diagram — Import Single Document



Chi tiết code

File: `service/knowledge/KnowledgeService.java` → `saveDocument()`

```
public List<KnowledgeDocument> saveDocument(String title, String content,
                                             String category, List<String> tags) {
    // ① Dedup check
    if (knowledgeDocumentRepository.existsByTitleIgnoreCase(title)) {
        throw new IllegalArgumentException("Document with title '" + title + "' already exists");
    }
}
```

```

// ② Chunking
List<String> chunks = textChunker.chunk(content);

if (chunks.size() <= 1) {
    // Single document – embed trực tiếp
    float[] embedding = embeddingService.embed(content);
    KnowledgeDocument doc = buildDocument(title, content, category, tags, embedding);
    return List.of(knowledgeDocumentRepository.save(doc));
}

// Chunked – embed batch + tạo nhiều records
List<float[]> embeddings = embeddingService.embedBatch(chunks);
List<KnowledgeDocument> savedDocs = new ArrayList<>();
for (int i = 0; i < chunks.size(); i++) {
    String chunkTitle = title + " [Part " + (i + 1) + "]";
    KnowledgeDocument doc = buildDocument(chunkTitle, chunks.get(i), category, tags, embeddings.get(i));
    savedDocs.add(knowledgeDocumentRepository.save(doc));
}
return savedDocs;
}

```

4.3 Text Chunking — Chi tiết

File: service/knowledge/TextChunker.java

Tham số

Param	Default	Mô tả
maxTokens	800	Tối đa tokens/chunk
overlapTokens	100	Tokens overlap giữa 2 chunk liền kề
TOKENS_PER_WORD	0.75	Ước lượng token/từ cho tiếng Việt

Thuật toán

- Ước lượng tokens = wordCount × 0.75
- Nếu ≤ 800 tokens → return [content] (không chunk)
- Nếu > 800 tokens:
 - Chia thành chunks dựa trên word count
 - Tìm natural break point (ưu tiên: paragraph > sentence > word)
 - Overlap 100 tokens (~133 từ) giữa các chunk liền kề

Natural Break Priority

- Priority 1: Paragraph break (\n\n)
 Priority 2: Sentence boundary (. ! ?)

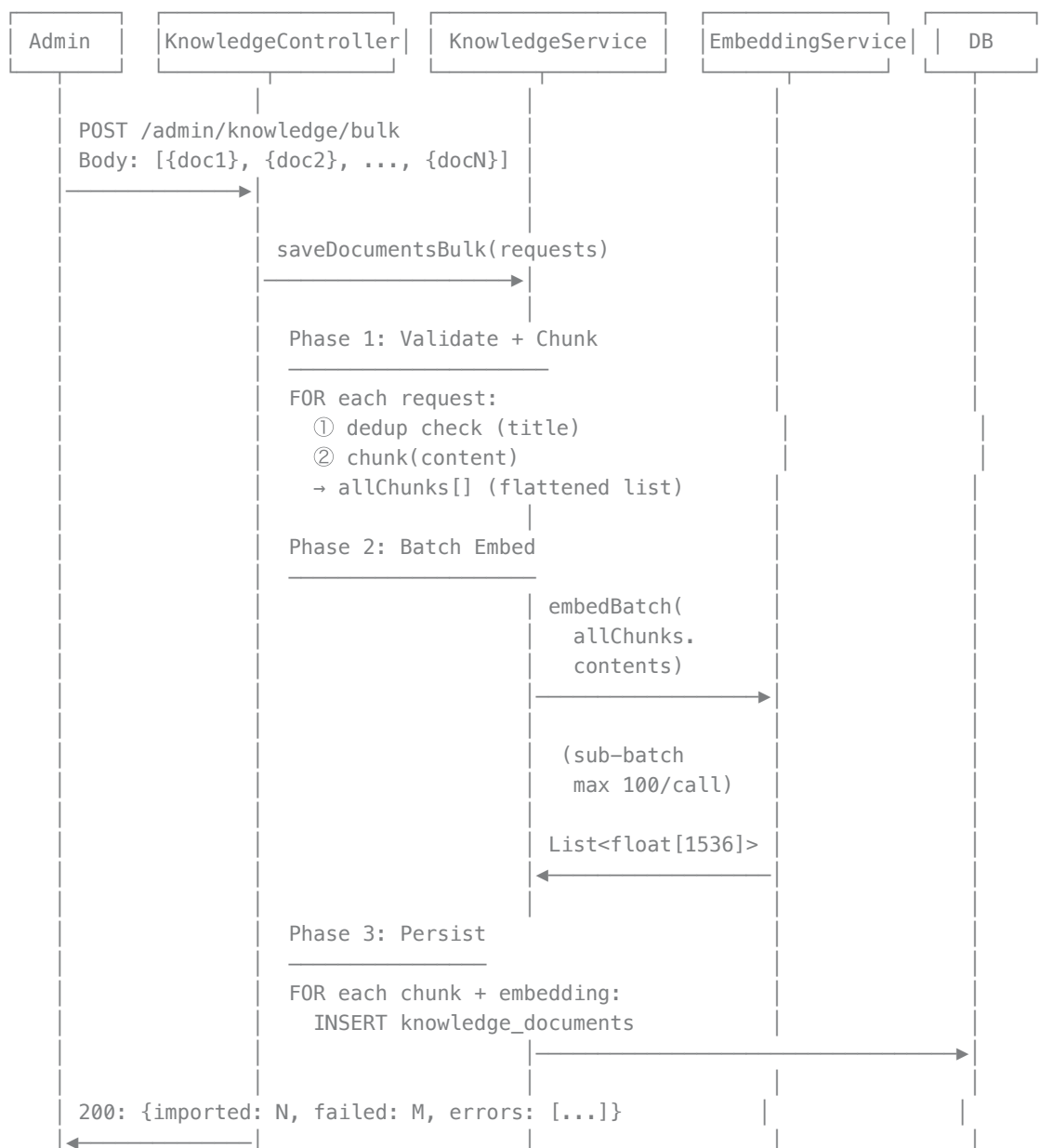
Ví dụ

Document: 1200 từ $\approx$ 900 tokens $>$ 800 threshold

→ Chunk 1: từ 1 – 1067 (800 tokens, tìm break gần nhất)

→ Chunk 2: từ 934 – 1200 (overlap 133 từ $\approx$ 100 tokens)

4.4 Sequence Diagram — Bulk Import



Batch Embedding — Sub-batching

File: service/knowledge/OpenAiEmbeddingService.java

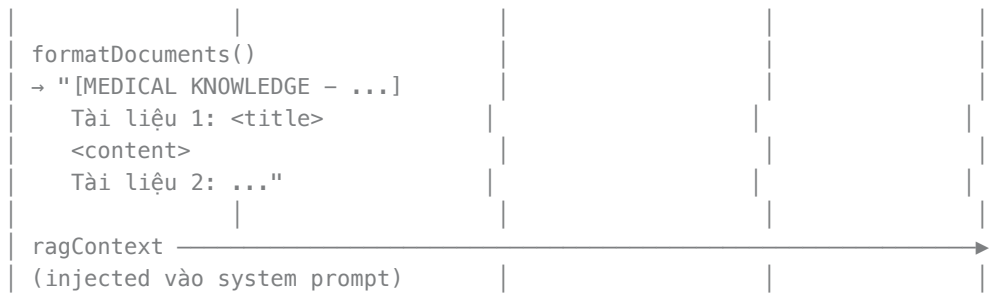
```
// OpenAI API limit: max 100 texts per call
for (int batchStart = 0; batchStart < validTexts.size(); batchStart += 100) {
    int batchEnd = Math.min(batchStart + 100, validTexts.size());
    List<String> subBatch = validTexts.subList(batchStart, batchEnd);
    List<float[]> subResult = callBatchApi(subBatch);
    validEmbeddings.addAll(subResult);
}
```

API call:

```
POST https://api.openai.com/v1/embeddings
{
  "input": ["text1", "text2", ...],
  "model": "text-embedding-3-small"
}
```

4.5 Sequence Diagram — Semantic Search (Runtime)





Graceful Fallback

```
// RagRetriever.retrieve() – nếu embedding API fail → return null
try {
    List<KnowledgeDocument> documents = knowledgeService.searchByText(...);
    if (documents.isEmpty()) return null;
    return formatDocuments(documents);
} catch (Exception e) {
    log.error("RAG retrieval failed, proceeding without RAG context");
    return null; // Chat vẫn hoạt động, chỉ không có RAG context
}
```

4.6 CRUD Operations – Update & Delete

Update Document

```
// KnowledgeService.updateDocument()
public KnowledgeDocument updateDocument(Long id, String title, String content,
                                         String category, List<String> tags) {
    KnowledgeDocument existing = findById(id);
    boolean contentChanged = !existing.getContent().equals(content);

    existing.setTitle(title);
    existing.setContent(content);
    existing.setCategory(category);
    existing.setTags(tags);

    if (contentChanged) {
        // Re-embed chỉ khi content thay đổi
        float[] newEmbedding = embeddingService.embed(content);
        existing.setEmbedding(newEmbedding);
    }
    return knowledgeDocumentRepository.save(existing);
}
```

Soft Delete

```
// KnowledgeService.softDelete() – is_active = false, KHÔNG xóa khỏi DB
public void softDelete(Long id) {
```

```
KnowledgeDocument document = findById(id);
document.setIsActive(false);
knowledgeDocumentRepository.save(document);
}
```

Document bị soft-delete sẽ **tự động bị loại khỏi search** vì query có `WHERE is_active = true` .

4.7 Tổng hợp APIs

Method	Path	Mô tả
POST	/api/v1/admin/knowledge	Tạo 1 document (auto-chunk + embed)
POST	/api/v1/admin/knowledge/bulk	Bulk import nhiều documents
GET	/api/v1/admin/knowledge?category=&page=&size=	List documents (phân trang)
GET	/api/v1/admin/knowledge/{id}	Lấy 1 document
PUT	/api/v1/admin/knowledge/{id}	Update document (auto re-embed)
DELETE	/api/v1/admin/knowledge/{id}	Soft delete
POST	/api/v1/admin/knowledge/search	Semantic search

File: 04_flow_rag_build.md | Project: CareTalk Backend